

significantly affect our results.

**Experiment settings.** We explore the impact of different temperatures on the validity of the presented results. We adopt the same setting in the transformation generalization.

**Findings.** As illustrated in Figure 14a, LLMs tend to generate consistent and reliable CoT reasoning across a broad range of temperature settings (e.g., from  $1e-5$  up to 1), provided the values remain within a suitable range. This stability is maintained even when the models are evaluated under a variety of distribution shifts.

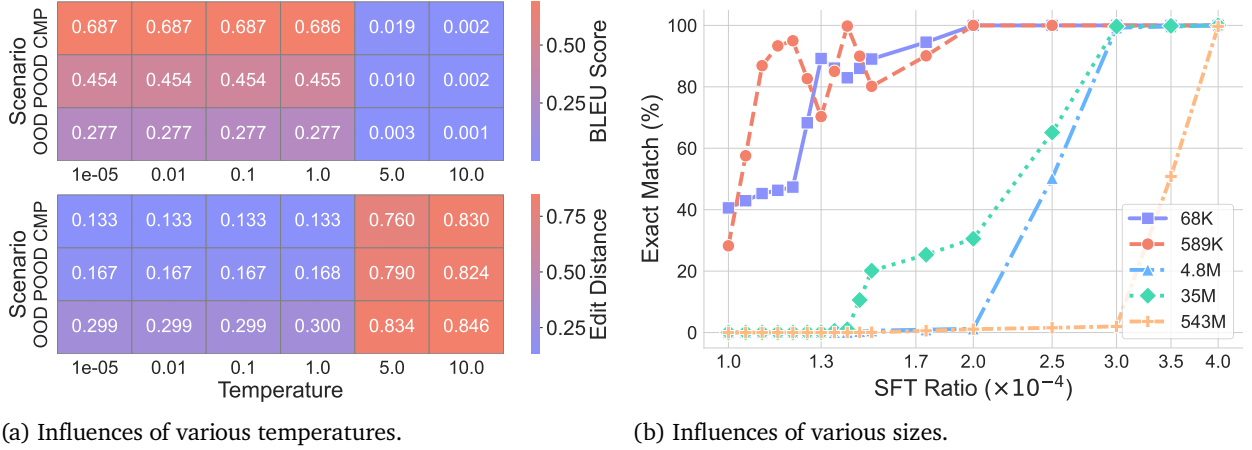


Figure 14 | Temperature and model size. The findings hold under different temperatures and model sizes.

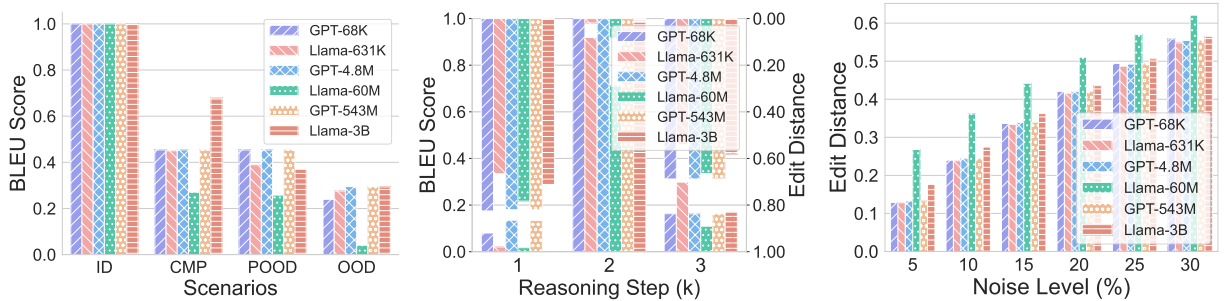


Figure 15 | Task, length, and format generalization of LLMs with settings. The data distribution lens is invariant across LLMs with various sizes and architectures.

Table 6 | Task generalization performance of SOTA LLMs (mean  $\pm$  std).

| Model              | Scenario | Exact Match (%)   | Edit Distance   | BLEU Score      |
|--------------------|----------|-------------------|-----------------|-----------------|
| LLaMA3-8B          | ID       | 100.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 | 1.00 $\pm$ 0.00 |
|                    | CMP      | 8.52 $\pm$ 0.00   | 0.23 $\pm$ 0.00 | 0.61 $\pm$ 0.00 |
|                    | POOD     | 0.00 $\pm$ 0.01   | 0.25 $\pm$ 0.01 | 0.46 $\pm$ 0.00 |
|                    | OOD      | 0.00 $\pm$ 0.00   | 0.27 $\pm$ 0.01 | 0.27 $\pm$ 0.00 |
| Qwen3-14B-Instruct | ID       | 100.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 | 1.00 $\pm$ 0.00 |
|                    | CMP      | 0.01 $\pm$ 0.01   | 0.17 $\pm$ 0.02 | 0.61 $\pm$ 0.00 |
|                    | POOD     | 0.00 $\pm$ 0.00   | 0.26 $\pm$ 0.01 | 0.42 $\pm$ 0.00 |
|                    | OOD      | 0.00 $\pm$ 0.00   | 0.38 $\pm$ 0.01 | 0.36 $\pm$ 0.00 |

**Experiment settings.** We further examine the influence of model size by employing the same

experimental configuration as used in the novel relation SFT study. In particular, we first pretrain models of different sizes using the transformation  $f_1 \circ f_1$ , and subsequently perform SFT on  $f_2 \circ f_2$  while varying the SFT ratios.

**Finding.** Fig. 14b shows the accuracy of models with different sizes using different SFT ratios, which closely matches the result of our default model size across all evaluated settings and configurations.

Table 7 | Reasoning step generalization of SOTA LLMs (mean  $\pm$  std).

| Model              | Reasoning Step ( $k$ ) | Exact Match (%)   | Edit Distance   | BLEU Score      |
|--------------------|------------------------|-------------------|-----------------|-----------------|
| LLaMA3-8B          | 1                      | 0.00 $\pm$ 0.00   | 0.75 $\pm$ 0.01 | 0.18 $\pm$ 0.00 |
|                    | 2                      | 100.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 | 1.00 $\pm$ 0.00 |
|                    | 3                      | 0.00 $\pm$ 0.00   | 0.54 $\pm$ 0.01 | 0.40 $\pm$ 0.00 |
| Qwen3-14B-Instruct | 1                      | 0.00 $\pm$ 0.00   | 0.54 $\pm$ 0.02 | 0.35 $\pm$ 0.00 |
|                    | 2                      | 100.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 | 1.00 $\pm$ 0.00 |
|                    | 3                      | 0.26 $\pm$ 0.05   | 0.65 $\pm$ 0.08 | 0.20 $\pm$ 0.00 |

Table 8 | Format generalization under different noise levels for SOTA LLMs (mean  $\pm$  std).

| Model              | Noise Level (%) | Exact Match (%)   | Edit Distance   | BLEU Score      |
|--------------------|-----------------|-------------------|-----------------|-----------------|
| LLaMA3-8B          | 0               | 100.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 | 1.00 $\pm$ 0.00 |
|                    | 5               | 32.30 $\pm$ 0.47  | 0.41 $\pm$ 0.02 | 0.04 $\pm$ 0.00 |
|                    | 10              | 17.74 $\pm$ 0.38  | 0.45 $\pm$ 0.00 | 0.04 $\pm$ 0.00 |
|                    | 15              | 9.27 $\pm$ 0.29   | 0.49 $\pm$ 0.05 | 0.04 $\pm$ 0.00 |
|                    | 20              | 4.72 $\pm$ 0.21   | 0.53 $\pm$ 0.03 | 0.03 $\pm$ 0.00 |
|                    | 25              | 2.32 $\pm$ 0.15   | 0.57 $\pm$ 0.00 | 0.03 $\pm$ 0.00 |
|                    | 30              | 1.09 $\pm$ 0.10   | 0.60 $\pm$ 0.08 | 0.03 $\pm$ 0.00 |
| Qwen3-14B-Instruct | 0               | 100.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 | 1.00 $\pm$ 0.00 |
|                    | 5               | 9.96 $\pm$ 0.30   | 0.22 $\pm$ 0.03 | 0.41 $\pm$ 0.00 |
|                    | 10              | 5.09 $\pm$ 0.22   | 0.35 $\pm$ 0.01 | 0.24 $\pm$ 0.00 |
|                    | 15              | 2.43 $\pm$ 0.15   | 0.44 $\pm$ 0.02 | 0.16 $\pm$ 0.00 |
|                    | 20              | 1.19 $\pm$ 0.11   | 0.52 $\pm$ 0.05 | 0.11 $\pm$ 0.00 |
|                    | 25              | 0.53 $\pm$ 0.07   | 0.57 $\pm$ 0.04 | 0.08 $\pm$ 0.00 |
|                    | 30              | 0.22 $\pm$ 0.05   | 0.62 $\pm$ 0.02 | 0.06 $\pm$ 0.00 |

## F.5. Internal Validity

Figures 8 and 15 illustrate task, length, and format generalization across a wide range of GPT- and LLaMA-style models. Increasing distribution discrepancy leads to a monotonic degradation of CoT performance regardless of architectural choice or parameter count. While larger models achieve uniformly higher absolute scores under near in-distribution conditions, they do not exhibit qualitatively different robustness profiles under moderate or severe shifts. This suggests that the observed failures of CoT reasoning cannot be attributed to insufficient capacity or architectural idiosyncrasies, but rather reflect a shared inductive bias learned from training distributions.

Model scaling consistently improves performance in ID and mildly shifted regimes (e.g., CMP or small reasoning-step extrapolation), but provides diminishing returns as the discrepancy increases toward POOD and OOD settings. In particular, larger models tend to preserve fluent intermediate reasoning traces even when final answers deteriorate, mirroring the same failure modes observed in smaller models. This pattern reinforces the interpretation that scaling primarily enhances pattern interpolation within the support of the training distribution, rather than enabling principled extrapolation beyond it.

Notably, the qualitative nature of errors remains stable across model sizes and architectures. Under task shifts, models consistently default to the closest seen transformation pattern; under length

shifts, they bias toward producing training-length reasoning chains; and under format perturbations, they remain sensitive to surface-level noise in structurally salient regions. The persistence of these behaviors across settings further supports the internal validity of the data distribution lens: CoT reasoning behaves as a distribution-sensitive generative process rather than an architecture-specific reasoning mechanism.

Taken together, these results demonstrate that our core findings are not artifacts of a particular model family, scale, or training instability. Instead, the dependence of CoT effectiveness on task, length, and format distributions emerges as a stable and reproducible phenomenon across controlled LLM instantiations. This strengthens the claim that distribution discrepancy—rather than model choice—is the dominant factor governing when CoT reasoning succeeds or fails.

## F.6. External Validity

As illustrated in Table 6, 7 & 8, the results on LLaMA3-8B-Instruct and Qwen3-14B-Instruct confirm that the behaviors identified in DATAALCHEMY persist in real-world, pretrained LLMs despite unknown and opaque training distributions. In the in-distribution (ID) setting, both models achieve perfect performance (100% exact match, zero edit distance, BLEU score of 1), indicating that the curated tasks are well within the expressive and optimization capacity of modern instruction-tuned models.

However, once distribution shifts are introduced, performance degrades sharply and systematically. Under composition (CMP), exact match accuracy drops to 8.52% for LLaMA3-8B and near zero (0.01%) for Qwen3-14B, while POOD and OOD settings result in complete failure (0% exact match) for both models. Correspondingly, edit distance increases and BLEU score decreases monotonically from CMP to OOD. These trends mirror those observed in models trained from scratch, reinforcing that the failure of CoT reasoning under task-level distribution shifts is not an artifact of synthetic training or limited model scale.

Reasoning-step generalization further highlights the limited extrapolation ability of CoT reasoning in SOTA models. Both LLaMA3-8B and Qwen3-14B achieve perfect performance at the in-distribution reasoning depth ( $k = 2$ ), but fail almost entirely at unseen depths. For  $k = 1$  and  $k = 3$ , exact match accuracy collapses to 0% in nearly all cases, accompanied by large edit distances and sharply reduced BLEU scores. Notably, Qwen3-14B exhibits a marginal non-zero accuracy (0.26%) at  $k = 3$ , but this gain is unstable and negligible relative to the ID performance. This pattern indicates that even large, instruction-tuned models do not acquire a length-agnostic or algorithmic reasoning procedure, but instead internalize a narrowly scoped reasoning template tied to the training distribution.

Format generalization experiments show consistent degradation trends across noise levels for both models. As noise increases from 0% to 30%, exact match accuracy decays smoothly from 100% to near zero, while edit distance increases and BLEU score decreases monotonically. Although LLaMA3-8B initially appears more sensitive at low noise levels (e.g., 32.30% exact match at 5% noise versus 9.96% for Qwen3-14B), both models converge to similarly poor performance under higher noise. The near-parallel degradation curves suggest that robustness to surface-level perturbations is constrained by distributional alignment.

Across all settings, the reported standard deviations are small relative to the absolute performance gaps between ID and shifted distributions. This indicates that the observed trends are stable across runs and not driven by sampling noise or stochastic decoding effects. In particular, zero or near-zero variance in ID settings confirms deterministic mastery of in-distribution patterns, while low variance under OOD conditions reflects consistently poor generalization rather than brittle or erratic behavior.

Overall, these results establish strong external validity for the proposed data distribution lens.